Response to the Editor

Manuscript PONE-D-26-28965R1

Zero-shot generalization of EEG attention-state decoding across increasingly stringent validation strategies on open datasets

Dear Dr. Jouzdani,

Thank you for your careful assessment and constructive comments. We have revised the manuscript to clarify the preprocessing and validation protocols, participant-level uncertainty, the interpretation of transfer comparisons and protocol-defined labels, and the ethical basis of this secondary analysis. The revisions to the Results section clarify the presentation and interpretation of the existing findings; no new analyses were conducted, and no experiments were rerun or metrics recalculated. Your comments are reproduced below and followed by our point-by-point responses.

Yours sincerely,

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Additional Editor Comments

Comment 1

Preprocessing and data leakage: Please specify exactly which data are used to fit ICA and all other data-dependent preprocessing steps in each validation protocol. Clarify whether any information from held-out participants or target datasets contributes to preprocessing, feature transformation, model selection, or other fitted operations. Explain how the protocol supports the claim of zero-shot generalization without target-specific adaptation.

Response

We thank the editor for raising this point. We clarified preprocessing and zero-shot evaluation in «Methodology» and «Validation strategies», and aligned «Data preparation» with those revisions.

In «Validation strategies», the opening paragraph now states the general rule for fitting ICA in the zero-shot protocols. Each of the four protocol descriptions then specifies which recordings were used for its ICA fits. The baseline description distinguishes the two datasets: ICA was fitted separately for each selected session in the mental attention-state dataset, whereas in SAM-40 it was fitted across a participant's selected trials and could span training and test folds. This shared decomposition may favor the SAM-40 baseline relative to the participant-disjoint protocols. The cross-subject, cross-task, and cross-dataset descriptions specify how ICA was fitted for held-out recordings and confirm that source and target recordings were never pooled in one fit.

In «Methodology», we clarify that ICA cleaning was the only fitted step that used unlabeled target recordings. Feature transformation was applied in the same form to source and target recordings; standardization, class weights, and decoder parameters were determined from source training data only. Target labels were never used, and target data played no role in model selection, hyperparameter search, early stopping, checkpoint selection, decoder retraining, or tuning. Thus, *zero-shot* refers to applying the source-trained decoder unchanged to target recordings, without target-guided decoder adaptation.

Comment 2

Uncertainty and sample size: Please discuss the limitations imposed by the five participants in Dataset A. Clarify the unit of resampling used for confidence intervals and ensure that the uncertainty estimates reflect the independent sampling units. Avoid interpreting small differences between baseline and cross-subject estimates as evidence of equivalence.

Response

Thank you for this comment. We revised «Generalization under increasingly stringent validation strategies» and «Performance consistency across decoders», both subsections of «Results and analysis», as well as «Discussion». These revisions clarify the confidence intervals, the interpretation of small differences between baseline and cross-subject validation, and the limitation imposed by the five participants in the mental attention-state dataset.

In «Generalization under increasingly stringent validation strategies», the third paragraph explains that the 95% intervals use 10,000 bootstrap resamples of target participants, with five participants in the mental attention-state dataset and 40 in SAM-40. The models are held fixed, and windows and folds are not resampled independently. The fifth paragraph reports the differences of 0.1 and 2.0 percentage points between baseline and cross-subject performance as descriptive and states that they do not establish equivalence.

In «Performance consistency across decoders», the first paragraph explains that the intervals shown across scenarios also use 10,000 resamples of target participants. Each resample uses the same participants for all five decoders.

In «Discussion», the fourth paragraph's second limitation explains that the mental attention-state dataset has only five participants. This limits population generalization and the precision of participant-level bootstrap intervals, despite the large number of EEG windows per participant.

Comment 3

Comparability of validation protocols: Please explain the extent to which the validation comparisons isolate changes in participant identity, task composition, and dataset. Ensure that transfer results are compared with appropriate target-specific baselines and that the conclusions do not attribute observed differences to a single factor when several factors change simultaneously.

Response

We thank the editor for raising this issue. We revised «Validation strategies», «Results and analysis», and «Discussion» to clarify what each validation comparison can show and how the transfer results should be interpreted.

In «Validation strategies», the second paragraph distinguishes the baseline from the three zero-shot protocols and explains what changes in each comparison. The baseline holds out recording units but may include participants represented in training. Cross-subject validation keeps the condition composition unchanged, making the unseen participant the main shift. Cross-task validation changes both the participant and task composition; comparisons involving the full SAM-40 composition also have active tasks in common. Cross-dataset validation changes the participant population, experimental paradigm, recording setup, and condition definitions together. The paragraph states that each zero-shot result is compared with the baseline for its target composition and that differences involving cross-task or cross-dataset transfer cannot be attributed to one factor alone.

In «Results and analysis», the sentence reporting the 6.4-point mean decrease now specifies that this summary includes both transfers between task-specific compositions and transfers involving the full SAM-40 composition, whose active tasks overlap with those of the task-specific compositions.

In «Discussion», we state explicitly that the observed reductions cannot be attributed to task composition or dataset alone because several factors changed together.

Comment 4

Interpretation of attention labels: Please maintain a clear distinction between classifying the protocol-defined conditions and decoding attention as a general psychological construct. Discuss the potential influence of task demands and other differences between resting and active cognitive conditions.

Response

We thank the editor for highlighting this distinction. We revised «Methodology» and «Discussion» to make clear that the analysis classifies protocol-defined conditions and to address differences between resting and active cognitive conditions.

In «Methodology», the second paragraph states that the high- and low-attention labels denote protocol-defined conditions rather than independently validated levels of attention, and that they may reflect differences in task demands, arousal, sensory input, stress, or motor responses between resting and active conditions. It specifies that the analyses therefore assess whether decoders discriminate the specified protocol conditions, not whether they decode attention as a general psychological construct.

In «Discussion», the fourth paragraph's first limitation now states that the reported accuracies reflect classification of the protocol-defined conditions rather than decoding of attention as a general psychological construct, and reiterates that task demands, stress, arousal, sensory input, or motor responses may also contribute, using the SAM-40 contrast of active tasks with rest while listening to music as an example.

Comment 5

Ethics and reporting: Please clarify the ethical approval, consent, and exemption status applicable to the original datasets and the present secondary analysis, as appropriate. Ensure that the ethics statement is consistent with the submission metadata.

Response

We thank the editor for this comment. In the text we added a new «Ethics statement» subsection under «Open-access datasets», extended the description of each dataset to report the consent and ethical oversight documented in the original studies, and revised the opening of the «Methodology» section to state explicitly that this study is a secondary analysis of two publicly available datasets.

For the mental attention-state dataset, the original study reported that participants were healthy university student volunteers who gave written informed consent after being informed of the study objectives and procedures, and that the study was conducted in accordance with the ethical guidelines of Mersin University. For SAM-40, the original study reported informed consent from all 40 participants and approval from the Institutional Ethics Committee of Gauhati University (GUIEC/2019/019, 15 October 2019), with voluntary and anonymous participation in compliance with the Declaration of Helsinki. Our «Ethics statement» makes clear that this work is a secondary analysis of these public recordings with no new data collection or participant contact, so no additional ethical approval or consent was required.

[TODO — supervisor to confirm.] Consistency between this statement and the submission metadata in Editorial Manager falls under the corresponding author's part of the submission, so we leave that confirmation to him.

Comment 6

Final consistency check: Please verify that the final clean manuscript, figures, supplementary tables, and response letter contain consistent terminology, numerical results, and methodological descriptions.

Response

Thank you for requesting a final consistency check. We reviewed the clean manuscript, figures, supplementary tables, and response letter. We aligned the descriptions of sampling, ICA fitting, SAM-40 recording units, model inputs, and the aggregation and comparison of transfer results, and checked terminology and numerical summaries across these materials.